



19. Which of the following is a type of architecture used in the computers nowadays?
 - a) Microarchitecture
 - b) Harvard Architecture
 - c) Von-Neumann Architecture
 - d) System Design
20. Which of the following is the subcategories of computer architecture?
 - a) Microarchitecture
 - b) Instruction set architecture
 - c) Systems design
 - d) All of the mentioned
21. Which of the architecture is power efficient?
 - a) RISC
 - b) ISA
 - c) IANA
 - d) CISC
22. What does CSA stands for?
 - a) Computer Service Architecture
 - b) Computer Speed Addition
 - c) Carry Save Addition
 - d) None of the mentioned
23. If an exception is raised and the succeeding instructions are executed completely, then the processor is said to have _____.
 - a) Generation word
 - b) Exception handling
 - c) Imprecise exceptions
 - d) None of the mentioned
24. To reduce the memory access time we generally make use of _____.
 - a) SDRAM's
 - b) Heaps
 - c) Cache's
 - d) Higher capacity RAM's
25. The IA-32 system follows which of the following design?
 - a) CISC
 - b) SIMD
 - c) RISC
 - d) None of the mentioned
26. Which of the following architecture is suitable for a wide range of data types?
 - a) IA-32
 - b) ARM
 - c) ASUS firebird
 - d) 68000
27. In IA-32 architecture along with the general flags, which of the following conditional flags are provided?
 - a) TF
 - b) IOPL
 - c) IF
 - d) All of the mentioned
28. The VLIW architecture follows _____ approach to achieve parallelism.
 - a) SISD
 - b) MIMD
 - c) MISD
 - d) SIMD
29. What does VLIW stands for?
 - a) Very Long Instruction Width
 - b) Very Large Instruction Word
 - c) Very Long Instruction Width
 - d) Very Long Instruction Word
30. In CISC architecture most of the complex instructions are stored in _____.
 - a) CMOS
 - b) Register
 - c) Transistors
 - d) Diodes
31. Both the CISC and RISC architectures have been developed to reduce the _____.
 - a) Time delay
 - b) Semantic gap
 - c) Cost
 - d) All of the mentioned
32. _____ are the different type/s of generating control signals.
 - a) Hardwired
 - b) Micro-instruction
 - c) Micro-programmed
 - d) Both Micro-programmed and Hardwired
33. If the instruction Add R1, R2, R3 is executed in a system which is pipelined, then the value of S is (Where S is term of the Basic performance equation).
 - a) 2
 - b) ~1
 - c) ~7
 - d) 2
34. The small extremely fast, RAM's all called as _____.
 - a) Heaps
 - b) Accumulators
 - c) Stacks
 - d) Cache
35. For a given FINITE number of instructions to be executed, which architecture of the processor provides for a faster execution?
 - a) ANSA
 - b) Super-scalar
 - c) ISA
 - d) All of the mentioned
36. What is the full form of ISA?
 - a) Industry Standard Architecture
 - b) International Standard Architecture
 - c) International American Standard
 - d) None of the mentioned
37. Which of the following is the fullform of CISC?
 - a) Complex Instruction Sequential Compilation
 - b) Complete Instruction Sequential Compilation
 - c) Computer Integrated Sequential Compiler
 - d) Complex Instruction Set Computer
38. The reason for the cells to lose their state over time is _____.
 - a) Use of Shift registers
 - b) The lower voltage levels
 - c) Usage of capacitors to store the charge
 - d) None of the mentioned
39. In order to read multiple bytes of a row at the same time, we make use of _____.
 - a) Memory extension
 - b) Cache
 - c) Shift register
 - d) Latch





40. The difference in the address and data connection between DRAM's and SDRAM's is _____
 - a) The requirement of more address lines in SDRAM's
 - b) The usage of a buffer in SDRAM's
 - c) The usage of more number of pins in SDRAM's
 - d) None of the mentioned
 41. The chip can be disabled or cut off from an external connection using _____
 - a) ACPT
 - b) RESET
 - c) LOCK
 - d) Chip select
 42. The controller multiplexes the addresses after getting the _____ signal.
 - a) INTR
 - b) ACK
 - c) RESET
 - d) Request
 43. The data is transferred over the RAMBUS as _____
 - a) Blocks
 - b) Swing voltages
 - c) Bits
 - d) Packets
 44. The memory devices which are similar to EEPROM but differ in the cost effectiveness is _____
 - a) CMOS
 - b) Memory sticks
 - c) Blue-ray devices
 - d) Flash memory
 45. The flash memory modules designed to replace the functioning of a hard disk is _____
 - a) RIMM
 - b) FIMM
 - c) Flash drives
 - d) DIMM
 46. The drawback of building a large memory with DRAM is _____
 - a) The Slow speed of operation
 - b) The large cost factor
 - c) The inefficient memory organisation
 - d) All of the mentioned
 47. In a 4K-bit chip organisation has a total of 19 external connections, then it has _____ address if 8 data lines are there.
 - a) 10
 - b) 12
 - c) 9
 - d) 8
 48. What does ISO stands for?
 - a) International Software Organisation
 - b) Industrial Software Organisation
 - c) International Standards Organisation
 - d) Industrial Standards Organisation
 49. The bit used to signify that the cache location is updated is _____
 - a) Flag bit
 - b) Reference bit
 - c) Update bit
 - d) Dirty bit
 50. During a write operation if the required block is not present in the cache then _____ occurs.
 - a) Write miss
 - b) Write latency
 - c) Write hit
 - d) Write delay
 51. While using the direct mapping technique, in a 16 bit system the higher order 5 bits are used for _____
 - a) Id
 - b) Word
 - c) Tag
 - d) Block
 52. The bit used to indicate whether the block was recently used or not is _____
 - a) Reference bit
 - b) Dirty bit
 - c) Control bit
 - d) Idol bit
 53. The number successful accesses to memory stated as a fraction is called as _____
 - a) Access rate
 - b) Success rate
 - c) Hit rate
 - d) Miss rate
 54. Who is the father of Computers?
 - a) James Gosling
 - b) Charles Babbage
 - c) Dennis Ritchie
 - d) Bjarne Stroustrup
 55. Which of the following is the correct abbreviation of COMPUTER?
 - a) Commonly Occupied Machines Used in Technical and Educational Research
 - b) Commonly Operated Machines Used in Technical and Environmental Research
 - c) Commonly Oriented Machines Used in Technical and Educational Research
 - d) Commonly Operated Machines Used in Technical and Educational Research
 56. Which of the following is the correct definition of Computer?
 - a) Computer is a machine or device that can be programmed to perform arithmetical or logic operation sequences automatically
 - b) Computer understands only binary language which is written in the form of 0s & 1s
 - c) Computer is a programmable electronic device that stores, retrieves, and processes the data
 - d) All of the mentioned
 57. What is the full form of CPU?
 - a) Computer Processing Unit
 - b) Computer Principle Unit
 - c) Central Processing Unit
 - d) Control Processing Unit
 58. Which of the following language does the computer understand?
 - a) Computer understands only C Language
 - b) Computer understands only Assembly Language
 - c) Computer understands only Binary Language
 - d) Computer understands only BASIC
 59. Which of the following computer language is written in binary codes only?
 - a) pascal
 - b) machine language
 - c) C
 - d) C#



60. Which of the following is the brain of the computer?
 - a) Central Processing Unit
 - b) Memory
 - c) Arithmetic and Logic unit
 - d) Control unit
61. Which of the following is not a characteristic of a computer?
 - a) Versatility
 - b) Accuracy
 - c) Diligence
 - d) I.Q.
62. Which of the following is the smallest unit of data in a computer?
 - a) Bit
 - b) KB
 - c) Nibble
 - d) Byte
63. Which of the following unit is responsible for converting the data received from the user into a computer understandable format?
 - a) Output Unit
 - b) Input Unit
 - c) Memory Unit
 - d) Arithmetic & Logic Unit
64. Which of the following monitor looks like a television and are normally used with non-portable computer systems?
 - a) LED
 - b) LCD
 - c) CRT
 - d) Flat Panel Monitors
65. Which of the following is not a type of computer code?
 - a) EDIC
 - b) ASCII
 - c) BCD
 - d) EBCDIC
66. Which of the following part of a processor contains the hardware necessary to perform all the operations required by a computer?
 - a) Controller
 - b) Registers
 - c) Cache
 - d) Data path
67. Which of the following is designed to control the operations of a computer?
 - a) User
 - b) Application Software
 - c) System Software
 - d) Utility Software
68. Which of the following device use positional notation to represent a decimal number?
 - a) Pascaline
 - b) Abacus
 - c) Computer
 - d) Calculator
69. Which of the following is used in EBCDIC?
 - a) Super Computers
 - b) Mainframes
 - c) Machine Codes
 - d) Programming
70. Which of the following are physical devices of a computer?
 - a) Hardware
 - b) Software
 - c) System Software
 - d) Package
71. Which of the following defines the assigned ordering among the characters used by the computer?
 - a) Accumulation
 - b) Sorting
 - c) Collating Sequence
 - d) Unicode
72. Which of the following storage is a system where a robotic arm will connect or disconnect off-line mass storage media according to the computer operating system demands?
 - a) Magnetic
 - b) Secondary
 - c) Virtual
 - d) Tertiary
73. Which of the following is known as the interval between the instant a computer makes a request for the transfer of data from a disk system to the primary storage and the instance the operation is completed?
 - a) Disk utilization time
 - b) Drive utilization time
 - c) Disk access time
 - d) Disk arrival time
74. Which of the following devices provides the communication between a computer and the outer world?
 - a) Compact
 - b) I/O
 - c) Drivers
 - d) Storage
75. Which of the following are the input devices that enable direct data entry into a computer system from source documents?
 - a) System Access devices
 - b) Data acquiring devices
 - c) Data retrieving devices
 - d) Data Scanning devices
76. Which of the following is the device used for converting maps, pictures, and drawings into digital form for storage in computers?
 - a) Image Scanner
 - b) Digitizer
 - c) MICR
 - d) Scanner
77. Which of the following can access the server?
 - a) Web Client
 - b) User
 - c) Web Browser
 - d) Web Server
78. Which of the following is known as the language made up of binary-coded instructions?
 - a) High level
 - b) BASIC
 - c) C
 - d) Machine
79. Which of the following package allows individuals to use personal computers for storing and retrieving their personal information?
 - a) Personal assistance package
 - b) Graphics package
 - c) Spreadsheet package
 - d) Animation package
80. Which of the following is created when a user opens an account in the computer system?
 - a) SFD
 - b) MFD
 - c) Subdirectory
 - d) RFD
81. Which of the following is a technique that marked the beginning of computer communications?
 - a) User Environment
 - b) Batch Environment
 - c) Time Sharing
 - d) Message passing





82. Which of the following is a type of technique in which dumb terminals are connected to a central computer system?
- Time Sharing
 - Message passing
 - Batch environment
 - User environment
83. Which of the following service allows a user to log in to another computer somewhere on the Internet?
- e-mail
 - UseNet
 - Telnet
 - FTP
84. Which of the following is not a type of computer on the basis of operation?
- Digital
 - Analog
 - Hybrid
 - Remote
85. Which of the following type of computer is mostly used for automatic operations?
- analog
 - digital
 - hybrid
 - remote
86. Which of the following invention gave birth to the much cheaper microcomputers?
- PDAs
 - Microprocessors
 - Microcomputers
 - Mainframes
87. Which of the following computers are lower than mainframe computers in terms of speed and storage capacity?
- Mainframes
 - Hybrid
 - Mini
 - Super
88. Which of the following is the first neural network computer?
- AN
 - AM
 - RFD
 - SNARC
89. Any signed negative binary number is recognised by its _____
- MSB
 - LSB
 - Byte
 - Nibble
90. The parameter through which 16 distinct values can be represented is known as _____
- Bit
 - Byte
 - Word
 - Nibble
91. If the decimal number is a fraction then its binary equivalent is obtained by _____ the number continuously by 2.
- Dividing
 - Multiplying
 - Adding
 - Subtracting
92. The representation of octal number (532.2)₈ in decimal is _____
- (346.25)₁₀
 - (532.864)₁₀
 - (340.67)₁₀
 - (531.668)₁₀
93. The decimal equivalent of the binary number (1011.011)₂ is _____
- (11.375)₁₀
 - (10.123)₁₀
 - (11.175)₁₀
 - (9.23)₁₀
94. An important drawback of binary system is _____
- It requires very large string of 1's and 0's to represent a decimal number
 - It requires sparingly small string of 1's and 0's to represent a decimal number
 - It requires large string of 1's and small string of 0's to represent a decimal number
 - It requires small string of 1's and large string of 0's to represent a decimal number
95. The decimal equivalent of the octal number (645)₈ is _____
- (450)₁₀
 - (451)₁₀
 - (421)₁₀
 - (501)₁₀
96. The largest two digit hexadecimal number is _____
- (FE)₁₆
 - (FD)₁₆
 - (FF)₁₆
 - (EF)₁₆
97. Representation of hexadecimal number (6DE)_H in decimal:
- $6 * 16^2 + 13 * 16^1 + 14 * 16^0$
 - $6 * 16^2 + 12 * 16^1 + 13 * 16^0$
 - $6 * 16^2 + 11 * 16^1 + 14 * 16^0$
 - $6 * 16^2 + 14 * 16^1 + 15 * 16^0$
98. The quantity of double word is _____
- 16 bits
 - 32 bits
 - 4 bits
 - 8 bits
99. The basic building blocks of the arithmetic unit in digital computers are _____
- Subtractors
 - Adders
 - Multiplexer
 - Comparator
100. All logic operations can be obtained by means of _____
- AND and NAND operations
 - OR and NOR operations
 - OR and NOT operations
 - NAND and NOR operations
101. The design of an ALU is based on _____
- Sequential logic
 - Combinational logic
 - Multiplexing
 - De-Multiplexing
102. If the two numbers are unsigned, the bit conditions of interest are the _____ carry and a possible _____ result.
- Input, zero
 - Output, one
 - Input, one
 - Output, zero





103. If the two numbers include a sign bit in the highest order position, the bit conditions of interest are the sign of the result, a zero indication and _____
a) An underflow condition
b) A neutral condition
c) An overflow condition
d) One indication
104. In parts of the processor, adders are used to calculate _____
a) Addresses
b) Table indices
c) Increment and decrement operators
d) All of the Mentioned
105. If A and B are the inputs of a half adder, the sum is given by _____
a) A AND B
b) A OR B
c) A XOR B
d) A EX-NOR B
106. If A and B are the inputs of a half adder, the carry is given by _____
a) A AND B
b) A OR B
c) A XOR B
d) A EX-NOR B
107. Half-adders have a major limitation in that they cannot _____
a) Accept a carry bit from a present stage
b) Accept a carry bit from a next stage
c) Accept a carry bit from a previous stage
d) Accept a carry bit from the following stages
108. If A, B and C are the inputs of a full adder then the carry is given by _____
a) A AND B OR (A OR B) AND C
b) A OR B OR (A AND B) C
c) (A AND B) OR (A AND B)C
d) A XOR B XOR (A XOR B) AND C
109. For subtracting 1 from 0, we use to take a _____ from neighbouring bits.
a) Carry
b) Borrow
c) Input
d) Output
110. How many outputs are required for the implementation of a subtractor?
a) 1
b) 2
c) 3
d) 4
111. Let the input of a subtractor is A and B then what the output will be if $A = B$?
a) 0
b) 1
c) A
d) B
112. Let A and B is the input of a subtractor then the borrow will be _____
a) A AND B'
b) A' AND B
c) A OR B
d) A AND B
113. The full subtractor can be implemented using _____
a) Two XOR and an OR gates
b) Two half subtractors and an OR gate
c) Two multiplexers and an AND gate
d) Two comparators and an AND gate
114. The output of a full subtractor is same as _____
a) Half adder
b) Full adder
c) Half subtractor
d) Decoder
115. The basic building blocks of the arithmetic unit in a digital computers are _____
a) Subtractors
b) Adders
c) Multiplexer
d) Comparator
116. The flag bits in an ALU is defined as _____
a) The total number of registers
b) The status bit conditions
c) The total number of control lines
d) All of the Mentioned
117. Looping on a K-map always results in the elimination of
a) Variables within the loop that appear only in their complemented form
b) Variables that remain unchanged within the loop
c) Variables within the loop that appear in both complemented and uncomplemented form
d) Variables within the loop that appear only in their uncomplemented form
118. Controlled inverter is also known as _____
a) Controlled buffer
b) NOT gate
c) Both controlled buffer and NOT gate
d) Controlled gate
119. Controlled buffers can be useful _____
a) To control the circuit's output into the bus
b) In comparison of component's output with its input
c) In increasing the output from its low input
d) All of the Mentioned
120. A logic circuit that provides a HIGH output for both inputs HIGH or both inputs LOW is _____
a) Ex-NOR gate
b) OR gate
c) Ex-OR gate
d) NAND gate
121. What is the major difference between half-adders and full-adders?
a) Full-adders are made up of two half-adders
b) Full adders can handle double-digit numbers
c) Full adders have a carry input capability
d) Half adders can handle only single-digit numbers





122. The binary subtraction of $0 - 0 = ?$
 a) Difference = 0, borrow = 0
 b) Difference = 1, borrow = 0
 c) Difference = 1, borrow = 1
 d) Difference = 0, borrow = 1
123. How many basic binary subtraction operations are possible?
 a) 1 b) 4 c) 3 d) 2

Digital Circuit

124. One way to make a four-bit adder to perform subtraction is by _____
 a) Inverting the output
 b) Inverting the carry-in
 c) Inverting the B inputs
 d) Grounding the B inputs
125. What distinguishes the look-ahead-carry adder?
 a) It is slower than the ripple-carry adder
 b) It is easier to implement logically than a full adder
 c) It is faster than a ripple-carry adder
 d) It requires advance knowledge of the final answer
126. What is one disadvantage of the ripple-carry adder?
 a) The interconnections are more complex
 b) More stages are required to a full adder
 c) It is slow due to propagation time
 d) All of the Mentioned

Digital Circuit

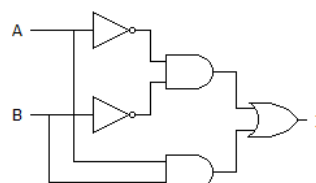
127. The inverter can be produced with how many NAND gates?
 a) 2 b) 1 c) 3 d) 4
128. How many NOT gates are required to implement the Boolean expression: $X = AB'C + A'BC$?
 a) 2 b) 3 c) 4 d) 5
129. What are carry generate combinations?
 a) If all the input are same then a carry is generated
 b) If all of the output are independent of the inputs
 c) If all of the input are dependent on the output
 d) If all of the output are dependent on the input
130. In serial addition, the addition is carried out _____
 a) 3 bit per second b) Byte by byte
 c) Bit by bit d) All bits at the same time
131. How many shift registers are used in a 4 bit serial adder?
 a) 4 b) 3 c) 2 d) 5
132. A D flip-flop is used in a 4-bit serial adder, why?
 a) It is used to invert the input of the full adder
 b) It is used to store the output of the full adder
 c) It is used to store the carry output of the full adder
 d) It is used to store the sum output of the full adder

BCD Adder

133. The decimal number system represents the decimal number in the form of _____
 a) Hexadecimal b) Binary coded
 c) Octal d) Decimal
134. 2^9 input circuit will have total of _____
 a) 32 entries b) 128 entries
 c) 256 entries d) 512 entries
135. BCD adder can be constructed with 3 IC packages each of _____
 a) 2 bits b) 3 bits c) 4 bits d) 5 bits
136. The output sum of two decimal digits can be represented in _____
 a) Gray Code b) Excess-3
 c) BCD d) Hexadecimal
137. The addition of two decimal digits in BCD can be done through _____
 a) BCD adder b) Full adder
 c) Ripple carry adder d) Carry look ahead
138. 3 bits full adder contains _____
 a) 3 combinational inputs
 b) 4 combinational inputs
 c) 6 combinational inputs
 d) 8 combinational inputs
139. Decimal digit in BCD can be represented by _____
 a) 1 input line b) 2 input lines
 c) 3 input lines d) 4 input lines
140. The number of logic gates and the way of their interconnections can be classified as _____
 a) Logical network
 b) System network
 c) Circuit network
 d) Gate network

Cobinational Circuits

141. Which of the following logic expressions represents the logic diagram shown?



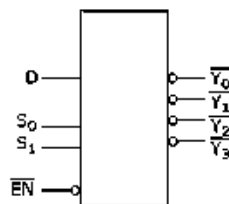
- a) $X = AB' + A'B$
 b) $X = (AB)' + AB$
 c) $X = (AB)' + A'B'$
 d) $X = A'B' + AB$





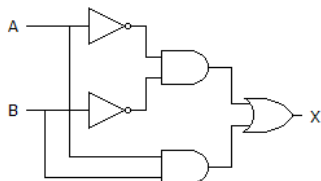
142. The device shown here is most likely a _____

- a) Comparator
- b) Multiplexer
- c) Inverter
- d) Demultiplexer



143. What type of logic circuit is represented by the figure shown below?

- a) XOR
- b) XNOR
- c) AND
- d) XAND



144. Which of the following combinations of logic gates can decode binary 1101?

- a) One 4-input AND gate
- b) One 4-input AND gate, one inverter
- c) One 4-input AND gate, one OR gate
- d) One 4-input NAND gate, one inverter

Multiplexers-1

145. What is a multiplexer?

- a) It is a type of decoder which decodes several inputs and gives one output
- b) A multiplexer is a device which converts many signals into one
- c) It takes one input and results into many output
- d) It is a type of encoder which decodes several inputs and gives one output

146. If the number of n selected input lines is equal to 2^n then it requires _____ select lines.

- a) 2
- b) m
- c) n
- d) $2n$

147. How many select lines would be required for an 8-line-to-1-line multiplexer?

- a) 2
- b) 4
- c) 8
- d) 3

148. A basic multiplexer principle can be demonstrated through the use of a _____

- a) Single-pole relay
- b) DPDT switch
- c) Rotary switch
- d) Linear stepper

Demultiplexers-2

149. In 1-to-4 demultiplexer, how many select lines are required?

- a) 2
- b) 3
- c) 4
- d) 5

150. In a multiplexer the output depends on its _____

- a) Data inputs
- b) Select inputs
- c) Select outputs
- d) Enable pin

151. How many select lines are required for a 1-to-8 demultiplexer?

- a) 2
- b) 3
- c) 4
- d) 5

152. How many AND gates are required for a 1-to-8 multiplexer?

- a) 2
- b) 6
- c) 8
- d) 5

153. Which IC is used for the implementation of 1-to-16 DEMUX?

- a) IC 74154
- b) IC 74155
- c) IC 74139
- d) IC 74138

Latch

154. Latch is a device with _____

- a) One stable state
- b) Two stable state
- c) Three stable state
- d) Infinite stable states

155. Why latches are called memory devices?

- a) It has capability to store 8 bits of data
- b) It has internal memory of 4 bit
- c) It can store one bit of data
- d) It can store infinite amount of data

156. How many types of latches are _____

- a) 4
- b) 3
- c) 2
- d) 5

157. The full form of SR is _____

- a) System rated
- b) Set reset
- c) Set ready
- d) Set Rated

158. The outputs of SR latch are _____

- a) x and y
- b) a and b
- c) s and r
- d) q and q'

159. The NAND latch works when both inputs are _____

- a) 1
- b) 0
- c) Inverted
- d) Don't cares

160. The first step of the analysis procedure of SR latch is to _____

- a) label inputs
- b) label outputs
- c) label states
- d) label tables

161. The inputs of SR latch are _____

- a) x and y
- b) a and b
- c) s and r
- d) j and k

162. When both inputs of SR latches are low, the latch _____

- a) Q output goes high
- b) Q' output goes high
- c) It remains in its previously set or reset state
- d) it goes to its next set or reset state

163. When both inputs of SR latches are high, the latch goes _____

- a) Unstable
- b) Stable
- c) Metastable
- d) Bistable





Flip-Flops-1

164. Latches constructed with NOR and NAND gates tend to remain in the latched condition due to which configuration feature?
- Low input voltages
 - Synchronous operation
 - Gate impedance
 - Cross coupling
165. The truth table for an S-R flip-flop has how many VALID entries?
- 1
 - 2
 - 3
 - 4
166. When both inputs of a J-K flip-flop cycle, the output will _____
- Be invalid
 - Change
 - Not change
 - Toggle
167. A basic S-R flip-flop can be constructed by cross-coupling of which basic logic gates?
- AND or OR gates
 - XOR or XNOR gates
 - NOR or NAND gates
 - AND or NOR gates
168. Whose operations are more faster among the following?
- Combinational circuits
 - Sequential circuits
 - Latches
 - Flip-flops
169. How many types of sequential circuits are?
- 2
 - 3
 - 4
 - 5
170. In S-R flip-flop, if $Q = 0$ the output is said to be _____
- Set
 - Reset
 - Previous state
 - Current state
171. The circuits of NOR based S-R latch classified as asynchronous sequential circuits, why?
- Because of inverted outputs
 - Because of triggering functionality
 - Because of cross-coupled connection
 - Because of inverted outputs & triggering functionality

Flip-Flops-2

172. What is an ambiguous condition in a NAND based S'-R' latch?
- $S'=0, R'=1$
 - $S'=1, R'=0$
 - $S'=1, R'=1$
 - $S'=0, R'=0$
173. The characteristic equation of S-R latch is _____
- $Q(n+1) = (S + Q(n))R'$
 - $Q(n+1) = SR + Q(n)R$
 - $Q(n+1) = S'R + Q(n)R$
 - $Q(n+1) = S'R + Q'(n)R$

174. The difference between a flip-flop & latch is _____
- Both are same
 - Flip-flop consist of an extra output
 - Latches has one input but flip-flop has two
 - Latch has two inputs but flip-flop has one
175. The S-R flip flop consist of _____
- 4 AND gates
 - Two additional AND gates
 - An additional clock input
 - 3 AND gates
176. What is the hold condition of a flip-flop?
- Both S and R inputs activated
 - No active S or R input
 - Only S is active
 - Only R is active
177. The circuit that is primarily responsible for certain flip-flops to be designated as edge-triggered is the _____
- Edge-detection circuit
 - NOR latch
 - NAND latch
 - Pulse-steering circuit

Flip-Flops-3

178. Which circuit is generated from D flip-flop due to addition of an inverter by causing reduction in the number of inputs?
- Gated JK-latch
 - Gated SR-latch
 - Gated T-latch
 - Gated D-latch
179. The characteristic of J-K flip-flop is similar to _____
- S-R flip-flop
 - D flip-flop
 - T flip-flop
 - Gated T flip-flop
180. A J-K flip-flop can be obtained from the clocked S-R flip-flop by augmenting _____
- Two AND gates
 - Two NAND gates
 - Two NOT gates
 - Two OR gates
181. How is a J-K flip-flop made to toggle?
- $J = 0, K = 0$
 - $J = 1, K = 0$
 - $J = 0, K = 1$
 - $J = 1, K = 1$
182. On a J-K flip-flop, when is the flip-flop in a hold condition?
- $J = 0, K = 0$
 - $J = 1, K = 0$
 - $J = 0, K = 1$
 - $J = 1, K = 1$
183. How many flip-flops are in the 7475 IC?
- 2
 - 1
 - 4
 - 8





D- Flip-Flops

184. In D flip-flop, D stands for _____
 a) Distant b) Data
 c) Desired d) Delay
185. The D flip-flop has _____ input.
 a) 1 b) 2 c) 3 d) 4
186. The D flip-flop has _____ output/outputs.
 a) 2 b) 3 c) 4 d) 1
187. A D flip-flop can be constructed from an _____ flip-flop.
 a) S-R b) J-K c) T d) S-K
188. In D flip-flop, if clock input is LOW, the D input _____
 a) Has no effect b) Goes high
 c) Goes low d) Has effect

Master-slave flip flops

189. The asynchronous input can be used to set the flip-flop to the _____.
 a) 1 state b) 0 state
 c) either 1 or 0 state d) forbidden State
190. Input clock of RS flip-flop is given to _____.
 a) Input b) Pulser
 c) Output d) Master slave flip-flop
191. D flip-flop is a circuit having _____.
 a) 2 NAND gates b) 3 NAND gates
 c) 4 NAND gates d) 5 NAND gates
192. In a positive edge triggered JK flip flop, a low J and low K produces?
 a) High state b) Low state
 c) Toggle state d) No Change State
193. If one wants to design a binary counter, the preferred type of flip-flop is _____.
 a) D type b) S-R type
 c) Latch d) J-K type
194. S-R type flip-flop can be converted into D type flip-flop if S is connected to R through _____.
 a) OR Gate b) AND Gate
 c) Inverter d) Full Adder
195. Which of the following flip-flops is free from the race around the problem?
 a) T flip-flop b) SR flip-flop
 c) Master-Slave Flip-flop d) D flip-flop
196. Which of the following is the Universal Flip-flop?
 a) S-R flip-flop b) J-K flip-flop
 c) Master slave flip-flop d) D Flip-flop
197. How many types of triggering take place in a flip flops?
 a) 3 b) 2 c) 4 d) 5

198. The S-R, J-K and D inputs are called _____.
 a) Asynchronous inputs b) Synchronous inputs
 c) Bidirectional inputs d) Unidirectional inputs
199. The circuit that generates a spike in response to a momentary change of input signal is called _____.
 a) R-C differentiator circuit
 b) L-R differentiator circuit
 c) R-C integrator circuit
 d) L-R integrator circuit

Flip-Flops realisation

200. To realise one flip-flop using another flip-flop along with a combinational circuit, known as _____.
 a) PREVIOUS state decoder
 b) NEXT state decoder
 c) MIDDLE state decoder
 d) PRESENT state decoder
201. For realisation of SR flip-flop from JK flip-flop, the excitation input will be obtained from _____.
 a) S and R b) R input
 c) J and K input d) D input
202. The K-map simplification for realisation of SR flip-flop from JK flip-flop is _____.
 a) $J=1, K=0$ b) $J=R, K=S$
 c) $J=S, K=R$ d) $J=0, K=1$
203. For realisation of D flip-flop from SR flip-flop, the external input is given through _____.
 a) S b) R
 c) D d) Both S and R
204. For D flip-flop to JK flip-flop, the characteristics equation is given by _____.
 a) $D=JQ(p)' + Q(p)K'$ b) $D=JQ(p)' + KQ(p)'$
 c) $D=JQ(p) + Q(p)K'$ d) $D=J'Q(p) + Q(p)K$

Counter

205. In digital logic, a counter is a device which _____.
 a) Counts the number of outputs
 b) Stores the number of times a particular event or process has occurred
 c) Stores the number of times a clock pulse rises and falls
 d) Counts the number of inputs
206. A counter circuit is usually constructed of _____.
 a) A number of latches connected in cascade form
 b) A number of NAND gates connected in cascade form
 c) A number of flip-flops connected in cascade
 d) A number of NOR gates connected in cascade form





207. What is the maximum possible range of bit-count specifically in n-bit binary counter consisting of 'n' number of flip-flops?

- a) 0 to 2^n b) 0 to 2^{n+1}
c) 0 to 2^{n-1} d) 0 to $2^{n+1/2}$

208. A decimal counter has _____ states.

- a) 5 b) 10 c) 15 d) 20

209. Ripple counters are also called _____

- a) SSI counters b) Asynchronous counters
c) Synchronous counters d) VLSI counters

210. BCD counter is also known as _____

- a) Parallel counter b) Decade counter
c) Synchronous counter d) VLSI counter

211. The parallel outputs of a counter circuit represent the _____

- a) Parallel data word b) Clock frequency
c) Counter modulus d) Clock count

Asynchronous Counter

212. How many natural states will there be in a 4-bit ripple counter?

- a) 4 b) 8 c) 16 d) 32

213. Internal propagation delay of asynchronous counter is removed by _____

- a) Ripple counter b) Ring counter
c) Modulus counter d) Synchronous counter

214. How many flip-flops are required to construct a decade counter?

- a) 4 b) 8 c) 5 d) 10

215. The terminal count of a typical modulus-10 binary counter is _____

- a) 0000 b) 1010
c) 1001 d) 1111

216. How many different states does a 3-bit asynchronous counter have?

- a) 2 b) 4 c) 8 d) 16

217. A ripple counter's speed is limited by the propagation delay of _____

- a) Each flip-flop
b) All flip-flops and gates
c) The flip-flops only with gates
d) Only circuit gates

Counter IC

218. How many different states does a 2-bit asynchronous counter have?

- a) 1 b) 4
c) 2 d) 8

219. Which one is a 4-bit binary ripple counter?

- a) IC 7493 b) IC 7490
c) IC 7491 d) IC 7492

220. In a 4-bit decade counter, four master-slave flip-flops are internally connected to provide a _____ bit counter.

- a) Divide-by-2 & divide-by-6
b) Divide-by-6 & divide-by-8
c) Divide-by-2 & divide-by-5
d) Divide-by-4 & divide-by-8

221. Ripple counter IC has _____

- a) 10 pins b) 11 pins
c) 12 pins d) 14 pins

Down Counter

222. Which of the following statements are true?

- a) Asynchronous events does not occur at the same time
b) Asynchronous events are controlled by a clock
c) Synchronous events does not need a clock to control them
d) Only asynchronous events need a control clock

223. A _____ down counter using n-flip-flops count _____

- a) Downward from a maximum count
b) Upward from a minimum count
c) Downward from a minimum to maximum count
d) Toggles between Up and Down count

224. How many different states does a 3-bit asynchronous down counter have?

- a) 2 b) 4
c) 6 d) 8

Ripple Counter

225. Modulus refers to _____

- a) A method used to fabricate decade counter units
b) The modulus of elasticity, or the ability of a circuit to be stretched from one mode to another
c) An input on a counter that is used to set the counter state, such as UP/DOWN
d) The maximum number of states in a counter sequence

226. A sequential circuit design is used to _____

- a) Count up b) Count down
c) Decode an end count d) Count in a random order

227. Which counters are often used whenever pulses are to be counted and the results displayed in decimal?

- a) Synchronous b) Bean
c) Decade d) BCD

228. Normally, the synchronous counter is designed using _____

- a) S-R flip-flops b) J-K flip-flops
c) D flip-flops d) T flip-flops





229. MOD-16 counter requires _____ no. of states.
a) 8 b) 4 c) 16 d) 32
230. High speed counter is _____
a) Ring counter b) Ripple counter
c) Synchronous counter d) Asynchronous counter
231. Program counter in a digital computer _____
a) Counts the number of programs run in the machine
b) Counts the number of times a subroutine
c) Counts the number of time the loops are executed
d) Points the memory address of the current or the next instruction
232. Fundamental mode is another name for _____
a) Level operation b) Pulse operation
c) Clock operation d) Edge operation

Counter Implementation

233. A 5-bit asynchronous binary counter is made up of five flip-flops, each with a 12 ns propagation delay. The total propagation delay ($t_{p(tot)}$) is _____
a) 12 ms b) 24 ns
c) 48 ns d) 60 ns
234. A ripple counter's speed is limited by the propagation delay of _____
a) Each flip-flop
b) All flip-flops and gates
c) The flip-flops only with gates
d) Only circuit gates
235. The main drawback of a ripple counter is that _____
a) It has a cumulative settling time
b) It has a distributive settling time
c) It has a productive settling time
d) It has an associative settling time

Registers

236. A register is defined as _____
a) The group of latches for storing one bit of information
b) The group of latches for storing n-bit of information
c) The group of flip-flops suitable for storing one bit of information
d) The group of flip-flops suitable for storing binary information
237. How many types of registers are?
a) 2 b) 3 c) 4 d) 5

238. The main difference between a register and a counter is _____
a) A register has no specific sequence of states
b) A counter has no specific sequence of states
c) A register has capability to store one bit of information but counter has n-bit
d) A register counts data
239. In D register, 'D' stands for _____
a) Delay b) Decrement
c) Data d) Decay
240. Registers capable of shifting in one direction is _____
a) Universal shift register
b) Unidirectional shift register
c) Unipolar shift register
d) Unique shift register
241. A register that is used to store binary information is called _____
a) Data register b) Binary register
c) Shift register d) D – Register
242. A shift register is defined as _____
a) The register capable of shifting information to another register
b) The register capable of shifting information either to the right or to the left
c) The register capable of shifting information to the right only
d) The register capable of shifting information to the left only
243. In serial shifting method, data shifting occurs _____
a) One bit at a time b) simultaneously
c) Two bit at a time d) Four bit at a time

Shift Registers

244. Based on how binary information is entered or shifted out, shift registers are classified into _____ categories.
a) 2 b) 3 c) 4 d) 5
245. A shift register that will accept a parallel input or a bidirectional serial load and internal shift features is called as?
a) Tristate b) End around
c) Universal d) Conversion
246. How can parallel data be taken out of a shift register simultaneously?
a) Use the Q output of the first FF
b) Use the Q output of the last FF
c) Tie all of the Q outputs together
d) Use the Q output of each FF





247. What is meant by the parallel load of a shift register?
- All FFs are preset with data
 - Each FF is loaded with data, one at a time
 - Parallel shifting of data
 - All FFs are set with data
248. An 8-bit serial in/serial out shift register is used with a clock frequency of 2 MHz to achieve a time delay (td) of _____
- 16 us
 - 8 us
 - 4 us
 - 2 us

Universal shift register

249. In a parallel in/parallel out shift register, $D_0 = 1$, $D_1 = 1$, $D_2 = 1$, and $D_3 = 0$. After three clock pulses, the data outputs are _____
- 1110
 - 0001
 - 1100
 - 1000
250. The group of bits 10110111 is serially shifted (right-most bit first) into an 8-bit parallel output shift register with an initial state 11110000. After two clock pulses, the register contains _____
- 10111000
 - 10110111
 - 11110000
 - 11111100
251. In a 4-bit Johnson counter sequence, there are a total of how many states or bit patterns?
- 1
 - 3
 - 4
 - 8
252. How much storage capacity does each stage in a shift register represent?
- One bit
 - Two bits
 - Four bits
 - Eight bits

Shift Register Counter

253. When is it important to use a three-state buffer?
- When two or more outputs are connected to the same input
 - When all outputs are normally HIGH
 - When all outputs are normally LOW
 - When two or more outputs are connected to two or more inputs
254. By adding recirculating lines to a 4-bit parallel-in serial-out shift register, it becomes a _____ and _____ out register.
- Parallel-in, serial, parallel
 - Serial-in, parallel, serial
 - Series-parallel-in, series, parallel
 - Bidirectional in, parallel, series

255. How much storage capacity does each stage in a shift register represent?
- One bit
 - Two bits
 - Four bits
 - Eight bits

Ring Counter

256. What is the difference between a shift-right register and a shift-left register?
- There is no difference
 - The direction of the shift
 - Propagation delay
 - The clock input
257. Which is not characteristic of a shift register?
- Serial in/parallel in
 - Serial in/parallel out
 - Parallel in/serial out
 - Parallel in/parallel out

Up down Counter

258. UP-DOWN counter is also known as _____
- Dual counter
 - Multi counter
 - Multimode counter
 - Two Counter
259. In an UP-counter, each flip-flop is triggered by _____
- The output of the next flip-flop
 - The normal output of the preceding flip-flop
 - The clock pulse of the previous flip-flop
 - The inverted output of the preceding flip-flop
260. In DOWN-counter, each flip-flop is triggered by _____
- The output of the next flip-flop
 - The normal output of the preceding flip-flop
 - The clock pulse of the previous flip-flop
 - The inverted output of the preceding flip-flop
261. Binary counter that count incrementally and decrement is called _____
- Up-down counter
 - LSI counters
 - Down counter
 - Up counter
262. In 4-bit up-down counter, how many flip-flops are required?
- 2
 - 3
 - 4
 - 5
263. A modulus-10 counter must have _____
- 10 flip-flops
 - 4 Flip-flops
 - 2 flip-flops
 - Synchronous clocking
264. A 4-bit binary up counter has an input clock frequency of 20 kHz. The frequency of the most significant bit is _____
- 1.25 kHz
 - 2.50 kHz
 - 160 kHz
 - 320 kHz





Memory

265. Memory is a/an _____
- Device to collect data from other computer
 - Block of data to keep data separately
 - Indispensable part of computer
 - Device to connect through all over the world
266. A register is able to hold _____
- Data
 - Word
 - Nibble
 - Both data and word
267. A register file holds _____
- A large number of word of information
 - A small number of word of information
 - A large number of programs
 - A modest number of words of information
268. The very first computer memory consisted of _____
- A small display
 - A large memory storage equipment
 - An automatic keyboard input
 - An automatic mouse input
269. Which one of the following has capability to store data in extremely high densities?
- Register
 - Capacitor
 - Semiconductor
 - Flip-Flop
270. A large memory is compressed into a small one by using _____
- LSI semiconductor
 - VLSI semiconductor
 - CDR semiconductor
 - SSI semiconductor
271. CD-ROM refers to _____
- Floppy disk
 - Compact Disk-Read Only Memory
 - Compressed Disk-Read Only Memory
 - Compressed Disk- Random Access Memory

Memory-2

272. ROM consist of _____
- NOR and OR arrays
 - NAND and NOR arrays
 - NAND and OR arrays
 - NOR and AND arrays
273. ASIC stands for _____
- Application Special Integrated Circuits
 - Applied Special Integrated Circuits
 - Application Specific Integrated Circuits
 - Applied Specific Integrated Circuits

Memory-3

274. Memories are classified into _____ categories.
- 3
 - 4
 - 5
 - 6
275. Secondary memory is also known as _____
- Registers
 - Main Memory
 - RAM
 - Both registers and main memory
276. In a computer, registers are present _____
- Within control unit
 - Within RAM
 - Within ROM
 - Within CPU
277. Which of the following has the lowest access time?
- RAM
 - ROM
 - Registers
 - Flag
278. Main memories of a computer, usually made up of _____
- Registers
 - Semiconductors
 - Counters
 - PLDs
279. A Random Access Memory is one in which _____
- Any location can be accessed sequentially
 - Any location can be accessed randomly
 - Any location can be accessed serially
 - Any location can be accessed parallelly
280. An example of RAM is _____
- Floppy disk
 - Hard disk
 - Magnetic tape memory
 - Semiconductor RAM
281. A static memory is one in which _____
- Content changes with time
 - Content doesn't changes with time
 - Memory is static always
 - Memory is dynamic always
282. A dynamic memory is one in which _____
- Content changes with time
 - Content doesn't changes with time
 - Memory is static always
 - Memory is dynamic always

Memory Device-4

283. To store 1-bit of information, how many transistor is/are used _____
- 1
 - 2
 - 3
 - 4
284. In dynamic memory, CCD stands for _____
- Charged Count Devices
 - Change Coupled Devices
 - Charge Coupled Devices
 - Charged Compact Disk





285. The example of non-volatile memory device is _____

- a) Magnetic Core Memory
- b) Read Only Memory
- c) Random Access Memory
- d) Both Magnetic Core Memory and Read Only Memory

Memory device 5

286. A memory is a collection of _____

- a) Unit cells
- b) Storage cells
- c) Data cells
- d) Binary cells

287. The data stored in a group of bits is called _____

- a) Nibble
- b) Word
- c) Byte
- d) Address

288. Each word consist of a sequence of _____

- a) Letters
- b) Binary numbers
- c) Hexadecimal numbers
- d) Gray codes

289. The capacity of a memory unit is _____

- a) The number of binary input stored
- b) The number of words stored
- c) The number of bytes stored
- d) All of the Mentioned

ROM-1

290. Which of the following has the capability to store the information permanently?

- a) RAM
- b) ROM
- c) Storage cells
- d) Both RAM and ROM

291. Since, ROM has the capability to read the information only then also it has been designed, why?

- a) For controlling purpose
- b) For loading purpose
- c) For booting purpose
- d) For erasing purpose

292. ROM is made up of _____

- a) NAND and OR gates
- b) NOR and decoder
- c) Decoder and OR gates
- d) NAND and decoder

293. Which is not a removable drive?

- a) Zip
- b) Hard disk
- c) Super Disk
- d) Jaz

294. In ROM, each bit combination that comes out of the output lines is called _____

- a) Memory unit
- b) Storage class
- c) Data word
- d) Address

ROM-3

295. IC 74187 is of _____

- a) 512 bits
- b) 1024 bits
- c) 256 bits
- d) 68 bits

296. Which of the following IC is of 256 bit?

- a) IC 74187
- b) IC 74189
- c) IC 74188
- d) IC 74186

297. How many address location a bipolar ROM has?

- a) 16
- b) 32
- c) 64
- d) 8

Programmable ROM-1

298. How many $8\text{ k} \times 1$ RAMs are required to achieve a memory with a word capacity of 8 k and a word length of eight bits?

- a) Eight
- b) Two
- c) One
- d) Four

RAM-1

299. What is the access time?

- a) The time taken to move a stored word from one bit to other bits after applying the address bits
- b) The time taken to write a word after applying the address bits
- c) The time taken to read a stored word after applying the address bits
- d) The time taken to erase a stored word after applying the address bits

300. Which of the following is not a type of memory?

- a) RAM
- b) FEPROM
- c) EEPROM
- d) ROM

301. RAM is also known as _____

- a) RWM
- b) MBR
- c) MAR
- d) ROM

302. If a RAM chip has n address input lines then it can access memory locations upto _____

- a) $2^{(n-1)}$
- b) $2^{(n+1)}$
- c) 2^n
- d) 22^n

RAM-2

303. The MOS technology based semiconductor ROMs are classified into _____ categories.

- a) 2
- b) 3
- c) 4
- d) 5

304. MOS ROM is constructed using _____

- a) FETs
- b) Transistors
- c) MOSFETs
- d) BJTs





305. Which ROM can be erased by an electrical signal?

- a) ROM b) Mask ROM
c) EPROM d) EEPROM

RAM-3

306. Dynamic RAM is more preferable than static RAM, why?

- a) DRAM is of the lowest cost, lowest density
b) DRAM is of the highest cost, reduced size
c) DRAM is of the lowest cost, highest density
d) DRAM is more flexible and lowest storage capacity

307. The memory size of DRAM is _____

- a) 1 to 100 MB b) 512 to 1024 MB
c) 64 to 512 MB d) 16 to 256 MB

308. DRAM uses of integrated MOS capacitors as instead of a flip-flop.

- a) Storage cell b) Memory cell
c) Dynamic cell d) Static cell

309. The main memory of a PC is made of _____

- a) Cache
b) Dynamic RAM
c) Static RAM
d) Both cache and dynamic RAM

310. Virtual memory consists of _____

- a) SRAM b) DRAM
c) Magnetic memory d) Main Memory

311. Cache memory acts between _____

- a) RAM and ROM b) CPU and RAM
c) CPU and Hard Disk d) CPU and ROM

RAM-4

312. RAMs are utilized in the computer as _____

- a) Scratch-pad b) Buffer
c) Main memory d) All of the Mentioned

313. Which one is self-compatible?

- a) ROM b) RAM
c) EROM d) PROM

ANSWER

1.	(b)	2.	(a)	3.	(d)	4.	(c)	5.	(c)
6.	(b)	7.	(d)	8.	(a)	9.	(a)	10.	(c)
11.	(a)	12.	(d)	13.	(c)	14.	(b)	15.	(a)
16.	(b)	17.	(a)	18.	(d)	19.	(c)	20.	(d)
21.	(a)	22.	(b)	23.	(c)	24.	(c)	25.	(a)
26.	(a)	27.	(d)	28.	(b)	29.	(d)	30.	(c)
31.	(b)	32.	(d)	33.	(b)	34.	(d)	35.	(b)
36.	(c)	37.	(d)	38.	(c)	39.	(d)	40.	(b)

41.	(d)	42.	(d)	43.	(b)	44.	(d)	45.	(c)
46.	(a)	47.	(c)	48.	(c)	49.	(d)	50.	(a)
51.	(c)	52.	(b)	53.	(c)	54.	(b)	55.	(d)
56.	(d)	57.	(c)	58.	(c)	59.	(b)	60.	(a)
61.	(d)	62.	(a)	63.	(b)	64.	(c)	65.	(a)
66.	(d)	67.	(c)	68.	(b)	69.	(b)	70.	(a)
71.	(c)	72.	(d)	73.	(d)	74.	(b)	75.	(d)
76.	(b)	77.	(a)	78.	(d)	79.	(a)	80.	(c)
81.	(c)	82.	(a)	83.	(c)	84.	(d)	85.	(c)
86.	(b)	87.	(c)	88.	(d)	89.	(a)	90.	(d)
91.	(b)	92.	(a)	93.	(a)	94.	(a)	95.	(c)
96.	(c)	97.	(a)	98.	(b)	99.	(b)	100.	(d)
101.	(b)	102.	(d)	103.	(c)	104.	(d)	105.	(c)
106.	(a)	107.	(c)	108.	(a)	109.	(b)	110.	(b)
111.	(a)	112.	(b)	113.	(b)	114.	(b)	115.	(b)
116.	(b)	117.	(c)	118.	(c)	119.	(a)	120.	(a)
121.	(c)	122.	(a)	123.	(b)	124.	(c)	125.	(c)
126.	(c)	127.	(b)	128.	(a)	129.	(b)	130.	(c)
131.	(c)	132.	(c)	133.	(b)	134.	(d)	135.	(c)
136.	(c)	137.	(a)	138.	(d)	139.	(d)	140.	(a)
141.	(d)	142.	(d)	143.	(b)	144.	(b)	145.	(b)
146.	(b)	147.	(d)	148.	(c)	149.	(a)	150.	(b)
151.	(b)	152.	(c)	153.	(a)	154.	(b)	155.	(c)
156.	(a)	157.	(b)	158.	(d)	159.	(a)	160.	(b)
161.	(c)	162.	(c)	163.	(c)	164.	(d)	165.	(c)
166.	(c)	167.	(c)	168.	(a)	169.	(a)	170.	(b)
171.	(c)	172.	(d)	173.	(a)	174.	(c)	175.	(b)
176.	(b)	177.	(a)	178.	(d)	179.	(a)	180.	(a)
181.	(d)	182.	(a)	183.	(c)	184.	(b)	185.	(a)
186.	(a)	187.	(a)	188.	(a)	189.	(c)	190.	(b)
191.	(c)	192.	(d)	193.	(d)	194.	(c)	195.	(a)
196.	(b)	197.	(a)	198.	(b)	199.	(a)	200.	(b)
201.	(c)	202.	(c)	203.	(c)	204.	(a)	205.	(b)
206.	(c)	207.	(c)	208.	(b)	209.	(b)	210.	(b)
211.	(d)	212.	(c)	213.	(d)	214.	(a)	215.	(c)
216.	(c)	217.	(a)	218.	(b)	219.	(a)	220.	(c)
221.	(d)	222.	(a)	223.	(a)	224.	(d)	225.	(d)
226.	(d)	227.	(d)	228.	(b)	229.	(c)	230.	(c)
231.	(d)	232.	(b)	233.	(d)	234.	(a)	235.	(a)
236.	(d)	237.	(c)	238.	(a)	239.	(c)	240.	(b)
241.	(b)	242.	(b)	243.	(a)	244.	(c)	245.	(c)
246.	(d)	247.	(a)	248.	(c)	249.	(a)	250.	(d)
251.	(a)	252.	(a)	253.	(a)	254.	(a)	255.	(a)
256.	(b)	257.	(a)	258.	(c)	259.	(b)	260.	(d)
261.	(a)	262.	(c)	263.	(b)	264.	(a)	265.	(c)
266.	(d)	267.	(d)	268.	(c)	269.	(b)	270.	(b)
271.	(b)	272.	(c)	273.	(a)	274.	(c)	275.	(d)
276.	(d)	277.	(b)	278.	(c)	279.	(b)	280.	(d)
281.	(c)	282.	(c)	283.	(a)	284.	(c)	285.	(d)
286.	(b)	287.	(b)	288.	(b)	289.	(c)	290.	(b)
291.	(c)	292.	(c)	293.	(b)	294.	(c)	295.	(b)
296.	(c)	297.	(b)	298.	(a)	299.	(c)	300.	(c)



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301.	(a)	302.	(c)	303.	(c)	304.	(d)	305.	(d)
306.	(c)	307.	(d)	308.	(b)	309.	(d)	310.	(a)
311.	(b)	312.	(d)	313.	(b)				



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